

# Performance Testing

| Field         | Details                                                  |
|---------------|----------------------------------------------------------|
| Date          | 30 September 2026                                        |
| Team ID       | SWTID-2026-3226                                          |
| Project Name  | FitBuddy – AI Fitness Plan Generator using Gemini Models |
| Maximum Marks | 5 Marks                                                  |

## Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. This document records the testing approach and available results for FitBuddy.

### Step 1: Testing Overview

| Field               | Details                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------|
| Testing Tool Used   | Manual browser testing and pytest (FastAPI TestClient); Postman planned for load testing   |
| Type of Testing     | Load-oriented API testing; stress testing after baseline validation.                       |
| Target Module / API | / (home page), /generate-workout (workout plan generation), /submit-feedback (plan update) |
| Test Environment    | Local FastAPI/Uvicorn environment (127.0.0.1:8000)                                         |
| Test Date           | 30 September 2026                                                                          |

### Step 2: Test Scenarios

| S.No | Test Scenario / Description                            | No. of Virtual Users | Duration (sec) | Expected Outcome                                                                   |
|------|--------------------------------------------------------|----------------------|----------------|------------------------------------------------------------------------------------|
| 1    | Home page load                                         | Planned: 1           | Planned        | Page loads without server error.                                                   |
| 2    | Workout plan generation endpoint (/generate-workout)   | Planned: 1           | Planned        | Valid fitness profile returns a 7-day plan and nutrition tip without server error. |
| 3    | Plan update with long feedback (up to 2000 characters) | Planned: 1           | Planned        | Updated plan returns without request failure.                                      |
| 4    | Different goal, intensity and experience level options | Planned: 1           | Planned        | Plan returns successfully for each selection.                                      |

### Step 3: Performance Test Results

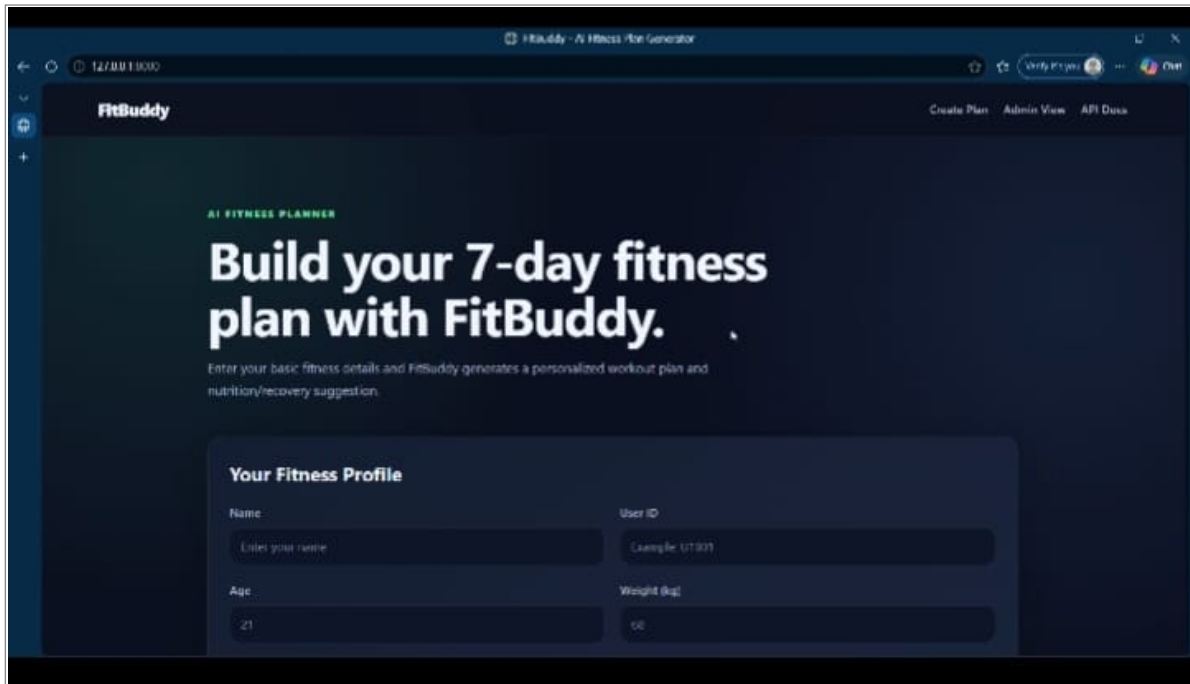
| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks                           |
|------|----------------------|--------------|--------------|----------------------|-----------------------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | Not measured | Pending execution    | No measured load result supplied. |
| 2    | Response Time (Max)  | < 5 seconds  | Not measured | Pending execution    | No measured load result supplied. |
| 3    | Throughput (Req/sec) | Baseline     | Not measured | Pending execution    | Requires concurrent requests.     |
| 4    | Error Rate           | < 1%         | Not measured | Pending execution    | Requires load-test run.           |
| 5    | CPU Utilization      | < 80%        | Not measured | Pending execution    | Requires monitoring during load.  |
| 6    | Memory Utilization   | < 80%        | Not measured | Pending execution    | Requires monitoring during load.  |

### Step 4: Observations & Analysis

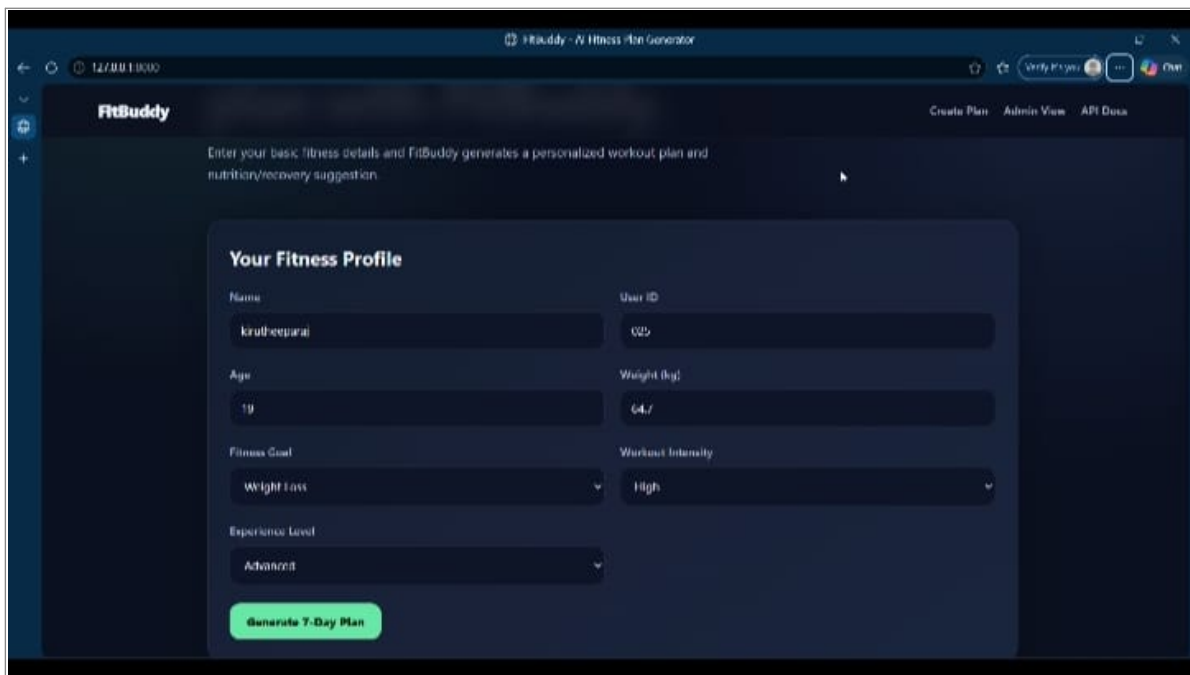
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Key Findings:</b></p> <p>FitBuddy exposes FastAPI endpoints for the home page, workout plan generation (/generate-workout), feedback-based plan update (/submit-feedback), an admin view and JSON API routes.</p> <p>In the local demo mode, a 7-day workout plan and nutrition tip were generated and displayed in the browser; live Gemini generation requires a GEMINI_API_KEY.</p> <p>Baseline performance values must be collected by repeated requests.</p> <p><b>Bottlenecks Identified:</b></p> <p>No measured bottleneck is claimed without an executed performance test.</p> <p><b>Optimization Steps Taken:</b></p> <p>Workout generation, nutrition tips and plan updates are handled by task-specific modules that share one cached Gemini client, and a built-in demo mode allows testing without live API calls.</p> |
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### Step 5: Screenshots / Evidence

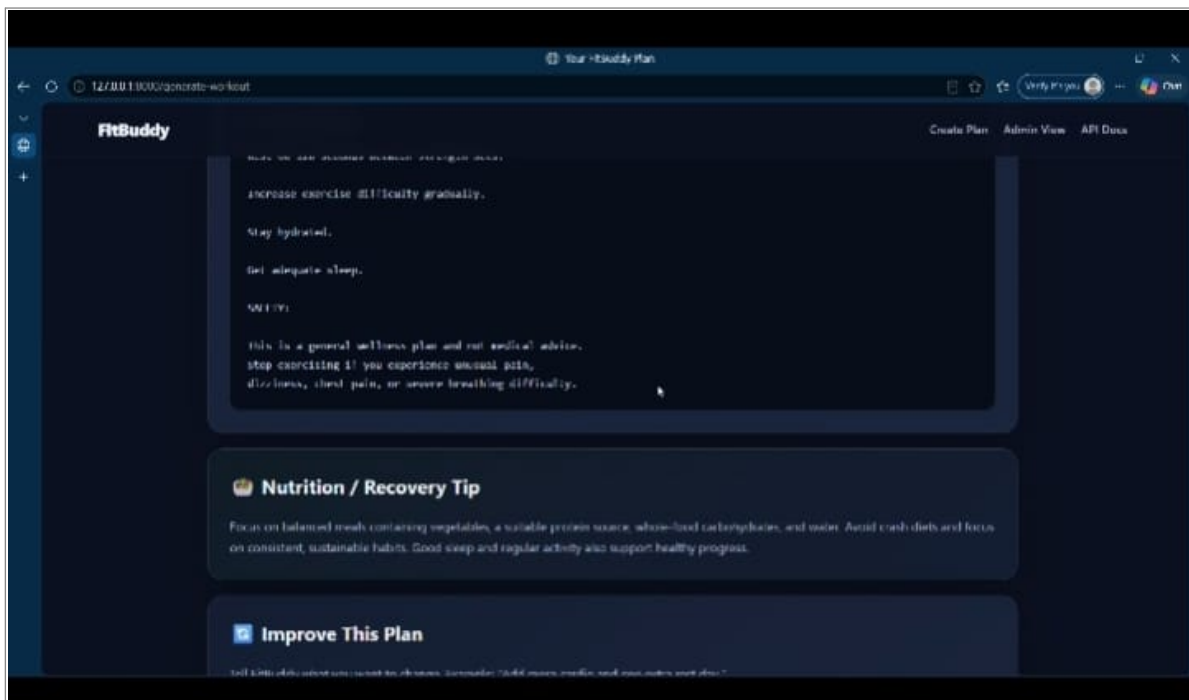
The screenshots below show the FitBuddy application running on the local FastAPI/Uvicorn environment. No load-test tool results are included because no load test has been executed yet.



Evidence 1 - FitBuddy landing page ("Build your 7-day fitness plan with FitBuddy.") running locally at 127.0.0.1:8000.



Evidence 2 - Fitness profile form with name, user ID, age, weight, fitness goal, workout intensity, experience level, and the "Generate 7-Day Plan" button.



Evidence 3 - Generated result at /generate-workout showing the workout plan safety notes, the Nutrition / Recovery Tip and the "Improve This Plan" feedback section.